Post-Disaster Damage Assessment and Rescue Guidance

CSE499B – Section 13, Group 5, Update 2

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Weekly Update:

Our segmentation model, a U-Net, labels every pixel of a post-disaster satellite image as background, intact building, damaged building or destroyed building. Ten training runs, each changing one thing in the recipe, all reached a mean intersection-over-union (mIoU: the overlap between predicted and true pixels, averaged over the four labels) of about 0.54, above our 0.52 target, but none held an IoU above 0.30 on destroyed buildings across two training seeds. The best, R9, which shows the model images of destroyed buildings more often and pastes extra destroyed buildings into training images, reached 0.300 on one seed and 0.294 on a second. Rather than keep tuning, we asked *why* destroyed buildings are missed, through four inexpensive tests. Each test's decision rule was written down before it ran, so that the result could not be bent to fit our expectations, and all four are scored on our 645 held-out validation image pairs. Alongside this, Ridita audited our data and built an offline guideline search, Aryan measured how much building counts help our vision-language model (a fine-tuned Qwen2.5-VL-7B that answers questions about the images), Noman built an agent that answers counting questions with SQL over a table of detected buildings, and Tamanna prototyped a pipeline that inspects each building separately.

Results Summary

Test	Question	Result	Decision
1. Error diagnosis	Why does the model miss destroyed buildings?	59.5% of the missed destroyed pixels are labelled background: the model fails to *find* rubble rather than to grade it	Give the next model the before-disaster image as well (R11)
2. One grade per building	Does a majority vote within each predicted building help?	Destroyed-building IoU fell from 0.292 to 0.228, and mIoU fell too	Rejected
3. Extra xBD images	Can we add more destroyed examples?	8,021 extra destroyed buildings, but 8,015 of them lie in images used by DisasterM3's question benchmark	Train on them for segmentation only (R12); R12 matched R11
4. SAM 3 as building finder	Can a general model find buildings better than ours?	On the before-image, cut into quarters: building IoU 0.630 against R9's 0.591, with no training on our data	SAM 3 finds the buildings in our find-then-grade model

Building IoU merges the three building classes into one and measures how well buildings are found at all. Removing the 40 validation pairs that duplicate training images changes none of these decisions (see the data check below).

Abrar Mohammed Tanzim Alam – Diagnosing the Destroyed-Building Gap

1. **Why destroyed buildings are missed:** we re-scored three trained models and traced where each missed destroyed pixel went. The rule, fixed in advance: if 40% or more are labelled background, the model does not recognise rubble as a building and should also see the before-disaster image, where the building still stands; otherwise it grades damage too mildly and needs more destroyed examples. The answer was 59.5% (58–60% for all three models), and in the other direction over a million background pixels were labelled destroyed: the model confuses rubble with bare ground. Turkey's 2023 earthquake alone holds 36.5% of all destroyed pixels, in 64 of the 645 pairs, so it dominates the score. The model that reads both images (R11) therefore ran first; it cut the share of its misses labelled background to 28% and raised mIoU from 0.537 to 0.581, but not its destroyed-building score.
2. **One grade per building:** giving each predicted building blob the class most of its pixels carry, a free post-processing step reported to help on the BRIGHT benchmark, lowered the destroyed-building score for every model (R9: 0.292 to 0.228) and lowered mIoU as well. Destroyed predictions are usually a minority within a blob and get outvoted; voting needs true building outlines, which the fourth test supplies.
3. **More destroyed examples from xBD:** xBD, the public dataset behind most DisasterM3 images, has images that DisasterM3 does not use for segmentation; 1,619 of them hold 8,021 destroyed buildings. We measured how DisasterM3 maps xBD's five damage labels onto its four (minor and major damage both become "damaged"; 97.8% of building pixels agree on the 1,792 images in both datasets) and converted them. However, 8,015 of those destroyed buildings lie in images used by DisasterM3's question benchmark, the multiple-choice questions on which our counting goal is scored. The model trained on them (R12) may therefore serve segmentation only and must never answer benchmark questions. R12 finished level with R11, consistent with the diagnosis that finding, not a shortage of examples, is the bottleneck.
4. **SAM 3 as the building finder:** SAM 3 (Meta, 2025), a general segmentation model prompted with the word "building" and used without any training, found buildings on the before-disaster image, cut into four quarters so that small buildings get more pixels, better than R9 (building IoU 0.630 against 0.591; on the uncut image it scored 0.484). It also counted them far more accurately (1.07 buildings found per true building, against 0.62, because R9 merges touching buildings into one blob). On the after-disaster image it covered only 9% of destroyed buildings, the same blindness to rubble, which independently confirms the diagnosis. Following its pre-set rule, SAM 3 became the finding stage of our find-then-grade model, in which a trained grader then assigns each building its damage level.

5. **A data check:** Ridita's data audit, confirmed by our own check, found that 40 of the 645 validation pairs share their before-image with a training pair: the same places, photographed for two Californian wildfires. A model can recognise those places from training, which flatters its score. Re-scored on the 605 clean pairs, every decision above holds: 57.7% of R9's misses are still labelled background, R9's best destroyed-building score moves from 0.300 to 0.287, and SAM 3's lead is unchanged (0.630 against 0.591). Two extra xBD images proved to be such duplicates too; they could only have helped R12, which still did not beat R11.

Ridita Afrin Riya – Data Audit & Offline Guideline Search

1. **DisasterM3 vs. BRIGHT optical audit:** found that 40 validation pairs (8.6% of the validation set's destroyed pixels) are near-copies of training pairs, which led to the E5 re-scoring. Showed that BRIGHT adds no destroyed examples for us (all its after-images are radar), that its Turkey, Morocco and Libya tiles must never enter training (they are the before-images of 66 validation pairs), and that 52% of before/after pairs are misaligned by more than 2 pixels.
2. **Offline guideline search:** built a local, cited search over 405 sections from six sources (EMS-98, INSARAG Vol. II and III, Sphere, xBD, IWG-SEM), combining MiniLM embeddings and BM25 keywords by reciprocal rank fusion, so the report writers can cite guidelines without internet access.

Aryan Sami – Segmentation-to-VLM Counting Study

1. **TEOChat-style grounding in the prompt:** following TEOChat (ICLR 2025), which found that a specialist U-Net localises buildings far better than a generalist VLM, built a pipeline that turns a segmentation mask into a building count plus normalised bounding boxes (0–1000) and injects them into the VLM's prompt.
2. **Counts in the prompt:** asked our fine-tuned 7B 1,703 optical counting questions with and without these findings. Given the correct count it scores 96.5%; from the image alone, 15.7%. The counting gap is therefore in *seeing* the buildings, not in reasoning about counts, which supports feeding segmenter counts to the VLM as text.

Abdullah Al Noman – SQL Counting Agent

1. **Text-to-SQL agent:** built the agent that answers counting questions by writing SQL over the per-building table; both tested models wrote valid SQL for all 100 sampled questions.
2. **Graded test (H2b):** ran the graded test of four small models (each in 4-bit and fp16) on 297 open questions. No model reached the 90% bar (best 73.9%), so the system will use fixed, pre-written queries.

Tamanna Akter Mou – Building-Crop Inspection Prototype

1. **Agentic vision pipeline:** prototyped a pipeline that crops each building at full resolution and asks the VLM about it, with a PaliGemma 2 baseline. A key lesson: when the VLM is shown the segmenter's class, it agrees almost every time, so agreement cannot be used as accuracy. Scores will be reported after a clean re-run against ground truth.

Next Week

- **Find-then-grade:** the results of the full model, in which a trained grader grades each building SAM 3 finds, including the destroyed-building target on the 605 clean pairs.
- **The vision-language model:** the result of its top-up training at full image resolution, and the start of its distillation into a smaller 3B model for field use.
- **Report system:** connect Ridita's offline guideline search to the DisasTeller report-writing agents, running on our own local models, and start the offline demo.